

Chapter 1

TT

1.1 CIC System

We first introduce some symbols in order to construct a formal language:

- ◊ $V = \{v_0, v_1, \dots\}$ is the set of term variables;
- ◊ $A = \{\alpha_0, \alpha_1, \dots\}$ is the set of type variables (below we take the union of A and V , and denote it by V);
- ◊ $C = \{\text{all UTF-8 strings}\}$ is the set of constants that are new definitions;
- ◊ $I = \{\text{Ind, Cstr, Elim}\}$ is the set of symbols used in defining inductions;
- ◊ $S = \{\text{Set, Prop}\} \cup \{\text{Type}_i\}_{i \geq 0}$ is the set of sorts;
- ◊ $P = \{., :, \equiv, (,)\}$ is the set of punctuation;
- ◊ $B = \{\lambda, \Pi\}$ is the set of binders.

Then we let

$$F = C|V|S|(BV : F.F)|(IF)|(FF)$$

Axioms $\mathcal{A} \subset S^2$ and rules $\mathcal{R} \subset S^3$ are defined as follows

$$\begin{aligned} \mathcal{A} &= \{(\text{Set}, \text{Type}_0), (\text{Prop}, \text{Type}_0), (\text{Type}_i, \text{Type}_{i+1})\} \\ \mathcal{R} &= \{(\text{Prop}, \text{Prop}, \text{Prop}), (\text{Set}, \text{Prop}, \text{Prop}), (\text{Type}_i, \text{Prop}, \text{Prop}), \\ &\quad (\text{Prop}, \text{Set}, \text{Set}), (\text{Set}, \text{Set}, \text{Set}), (\text{Type}_i, \text{Type}_j, \text{Type}_{\max\{i,j\}})\}. \end{aligned}$$

Note that $\Pi(\mathbf{x} : \mathbf{A})\varphi$ is the abbreviation of $\Pi x_0 : A_0. \Pi x_1 : A_1 \cdots \Pi x_m : A_m. \varphi$. $\mathbf{x} : \mathbf{A}$ is the abbreviation of $x_0 : A_0 x_1 : A_1 \cdots x_m : A_m$.

Definition 1.1.1 *The Arity has the form*

$$A = S|(\Pi V : F.A)$$

For any $X \in C$, the Strictly Positive Types has the form

$$P_X = X|(P_X \bigsqcup_{X \notin F})|(\Pi V : \bigsqcup_{X \notin F} .P_X)$$

The Constructors has the form

$$C_X = X|(C_X \bigsqcup_{X \notin F})|(\Pi V : \bigsqcup_{X \notin F} .C_X)|(\Pi V : P_X. \bigsqcup_{V \notin C_X})$$

Any element $C_X \setminus P_X$ is called recursive.

Definition 1.1.2 A definition D has the form

$$\mathbf{x} : \mathbf{A} \quad c\mathbf{x} \equiv d(\mathbf{x}) : a$$

note that $d(\mathbf{x})$ refers $d \in \mathbf{F}$ and $\text{Fr } d \subset \mathbf{x}$, that $x_i \in V, c \in C$ and that all A_i, d, a are in $\mathbf{F}$.

A environment Δ has the form $D_0 D_1 \cdots D_n$.

Definition 1.1.3 A judgment has the form

$$\Delta \quad \mathbf{x} : \mathbf{A} \quad \varphi : a$$

Definition 1.1.4 Given $\Delta = D_0 D_1 \cdots D_n$, $R_\Delta \subset \mathbf{F}^2$ is the smallest relation satisfies the following properties:

- ◇ contains $\{(c\mathbf{U}, d(\mathbf{U})) : \mathbf{x} : \mathbf{A} \quad c\mathbf{x} \equiv d(\mathbf{x}) : a \in D \in \Delta \text{ and } \mathbf{U} \in \mathbf{F}^*\}$;
- ◇ closed under the following operations: $\mathbf{F}-, -\mathbf{F}, \lambda V : \mathbf{F}-. , Cp_1 - p_2$.

Definition 1.1.5 R_β is the smallest relation satisfies

- ◇ contains $\{((\lambda x : a.\varphi)\chi, \varphi_x^X) : x \in V \text{ and all } \varphi, \chi, a \in \mathbf{F}\}$;
- ◇ closed under the following operations: $\mathbf{F}-, -\mathbf{F}, BV : \mathbf{F}-. , BV : -.\mathbf{F}, Cp_1 - p_2$.

Proposition 1.1.6 Any element in C_X and P_X has the form

$$\Pi(\mathbf{x} : \mathbf{M}).(XN)$$

which is unique **under** $\sim_{\beta, \Delta}$.

Suppose X is a constant symbol, $c_i \in C_X$, and $\Pi(\mathbf{x} : \mathbf{A}).s \in \mathbf{A}$, then

$$\begin{array}{c} \Delta \quad \Gamma \quad X : \Pi(\mathbf{x} : \mathbf{A}).s \quad c_0 : s \\ \vdots \\ \Delta \quad \Gamma \quad X : \Pi(\mathbf{x} : \mathbf{A}).s \quad c_m : s \\ \Delta \quad \Gamma \quad \text{Ind}X : \Pi(\mathbf{x} : \mathbf{A}).s \end{array}$$

and it immediately follows that

$$\begin{array}{c} \Delta \quad \Gamma \quad \text{Ind}X : \Pi(\mathbf{x} : \mathbf{A}).s \\ \Delta \quad \Gamma \quad \text{CstrInd}Xi : c_i \frac{\text{Ind}X}{X} \end{array}$$

(i is a constant belongs to C)

Assume that

- ◇ $\mathbf{F} \ni X : \Pi(\mathbf{x} : \mathbf{A}).s$
- ◇ $\mathbf{C} \ni Y : \Pi(\mathbf{x} : \mathbf{A}).(X\mathbf{x}) \rightarrow s'$
- ◇ $\mathbf{F} \ni d : c \in C_X$
- ◇ $P = \Pi(\mathbf{x}' : \mathbf{P}').(X\mathbf{m}') \in P_X$
- ◇ $p \in V$ is a new variable

we can define a new function $E_{X,Y,d} : C_X \rightarrow C_Y$ inductively, as follows

$$\begin{aligned} E_{X,Y,d}(X) &= Y \\ E_{X,Y,d}(cM) &= (E_{X,Y,d}(c)M)d \\ E_{X,Y,d}(P \rightarrow c) &= \Pi p : P.(\Pi(\mathbf{x}' : \mathbf{P}').Y\mathbf{m}'(p\mathbf{x}') \rightarrow E_{X,Y,dp}(c)) \\ E_{X,Y,d}(\Pi x : M.c) &= \Pi x : M.E_{X,Y,dx}(c) \end{aligned}$$

If $f_0, \dots, f_m$ and G are new constant symbols, we have

$$\begin{array}{lcl}
\Delta & \Gamma & G : \text{Ind}X \mathbf{a} \\
\Delta & \Gamma & Y : \Pi(\mathbf{x} : \mathbf{A}).((\text{Ind}X \mathbf{x}) \rightarrow s') \\
\Delta & \Gamma & f_0 : E_{\text{Ind}X, Y, \text{CstrInd}X0}(c_0) \\
& & \vdots \\
\Delta & \Gamma & f_m : E_{\text{Ind}X, Y, \text{CstrInd}Xm}(c_m) \\
\Delta & \Gamma & \text{Elim}GY f_0 f_1 \dots f_m : (Y \mathbf{a} G)
\end{array}$$

Again, suppose $u, v \in \mathbf{F}$ we define a function $H_{X,u,v} : \mathbf{C}_X \rightarrow \mathbf{F}$, by letting

$$\begin{aligned}
H_{X,u,v}(X) &= v \\
H_{X,u,v}(cM) &= H_{X,u,v}(c) \\
H_{X,u,v}(P \rightarrow c) &= \lambda p : P. H_{X,u,(vp\lambda(\mathbf{x}:\mathbf{P}). (vm(p\mathbf{x})))}(c) \\
H_{X,u,v}(\Pi x : M. c) &= \lambda x : M. H_{X,u,vx}(c)
\end{aligned}$$

Definition 1.1.7 Suppose $\mathbf{f}$ has the shape of m , $\rightarrow_\iota$ is a relation in $\mathbf{F}^2$ contains

$$\text{ElimCstrInd}Xi\mathbf{m}Y\mathbf{f} \rightarrow_\iota H_{\text{Ind}X, \lambda(\mathbf{x}:\mathbf{A}). \lambda G:\text{Ind}X\mathbf{x}. \text{Elim}GY\mathbf{f}, f_i}(c_i)\mathbf{m}$$

In COC, we have introduced the following sequent rules.

(Sort)

If $(s_1, s_2) \in \mathcal{A}$, then

$$- \quad - \quad s_1; s_2$$

(Var)

If $x \in V \setminus \mathbf{x}$, then

$$\begin{array}{lcl}
\Delta & \Gamma & a : s \\
\Delta & \Gamma & x : a \quad x : a
\end{array}$$

(Weak)

If $x \in V \setminus \mathbf{x}$, then

$$\begin{array}{lcl}
\Delta & \Gamma & \varphi : a \\
\Delta & \Gamma & b : s \\
\Delta & \Gamma & x : b \quad \varphi : a
\end{array}$$

(Form)

If $(s_1, s_2, s_3) \in \mathcal{R}$, then

$$\begin{array}{lcl}
\Delta & \Gamma & a_1 : s_1 \\
\Delta & \Gamma & x : a_1 \quad a_2 : s_2 \\
\Delta & \Gamma & \Pi x : a_1. a_2 : s_3
\end{array}$$

(Appl)

$$\begin{array}{lcl}
\Delta & \Gamma & \varphi : \Pi x : a. b \\
\Delta & \Gamma & \chi : a \\
\Delta & \Gamma & \varphi\chi : b_x^\chi
\end{array}$$

(Abst)

$$\begin{array}{lcl}
\Delta & \Gamma & x : a \quad \varphi : b \\
\Delta & \Gamma & \Pi x : a. b : s \\
\Delta & \Gamma & (\lambda x : a. \varphi) : (\Pi x : a. b)
\end{array}$$

(Def)

$$\begin{array}{lll} \Delta & \Gamma & \varphi : a \\ \Delta & \mathbf{y} : B & d(\mathbf{y}) : b \\ \Delta D & \Gamma & \varphi : a \end{array}$$

where D is the string

$$\mathbf{y} : B \quad c\mathbf{y} \equiv d(\mathbf{y}) : b$$

that $d(\mathbf{y})$ could be an empty word.

(Inst)

If Δ contains the element

$$\mathbf{y} : B \quad c\mathbf{y} \equiv d(\mathbf{y}) : a$$

that $d(\mathbf{y})$ could be empty, and we have $\mathbf{U} \in \mathbf{F}^*$ with the same size of $\mathbf{y}$, then

$$\begin{array}{lll} \Delta & \Gamma & s_1 : s_2 \ (\in \mathcal{A}) \\ \Delta & \Gamma & U_0 : B_0 \frac{\mathbf{U}}{\mathbf{y}} \\ & & \vdots \\ \Delta & \Gamma & U_m : B_m \frac{\mathbf{U}}{\mathbf{y}} \\ \Delta & \Gamma & c\mathbf{U} : a \frac{\mathbf{U}}{\mathbf{y}} \end{array}$$

(Conv)

If $a \sim_{\Delta, \beta} a'$, then

$$\begin{array}{lll} \Delta & \Gamma & \varphi : a \\ \Delta & \Gamma & a' : s \\ \Delta & \Gamma & \varphi : a' \end{array}$$